An Adaptive Skin Model and Its Application to Objectionable Image Filtering

ABSTRACT

We propose an adaptive skin-detection method, which allows modeling and detection of the true skin-color pixels with significantly higher accuracy and flexibility than previous methods. In principle, the proposed approach follows a two-step process. For a given image, we first perform a rough skin classification using a generic skin-model which defines the Skin-Similar space. The Skin-Similar space often contains many non-skin pixels due to the inevitable overlap in the color space between skin pixels and some non-skin pixels under the generic skin-model. The objective of the second step is to reduce the false-positive rate by analyzing the image under consideration. Specifically, in the second step, a Gaussian Mixture Model (GMM), specific to the image under consideration and refined from its Skin-Similar space, is derived using the standard Expectation-Maximization (EM) algorithm. We then use a Support Vector Machine (SVM) classifier to identify the skin Gaussian from the trained GMM by incorporating spatial and shape information of the skin pixels. Moreover, we examine how the improvement on skin detection by this adaptive skin-model impacts the detection accuracy in the application of Objectionable Image Filtering. We further propose a two-level classification scheme based on hierarchical bagging to improve the accuracy. Results of extensive experiments on large databases demonstrate the effectiveness and benefits of our adaptive skin-model.

Keywords

Skin Detection, Objectionable Image Filtering, Gaussian Mixture Model, Expectation Maximization, SVM

1. INTRODUCTION

Color usually presents a strong and intuitive cue in images with complex scenes. In recent years, skin detection has emerged as an active research topic in several practical applications, including human body detection [1, 2, 3], face tracking [4, 5, 6], and objectionable-image filtering [7, 8, 9].

Researchers have been studying various generic skin-models in a number of color spaces [10, 11]. However, we can expect variations when images are photographed in various settings, with different kinds of camera hardware, and under a wide range of lighting conditions. Moreover, ethnic groups present a range of skin tones that defy simplistic classification. Therefore, a generic skin-model is clearly inadequate to accurately capture the skin-colors in individual images. To improve the detection accuracy and reduce the false positive rate, one solution is to adopt an adaptive skin-model instead of a static one. Specific to the application of face-tracking in videos, an effective idea is to adapt skin models to the current frame by incorporating the information from previous frames. An interesting study was reported in [6] to explain the skin model transformation in consecutive frames with an affine motion model. Under this affine motion assumption, the authors propose a second-order Markov model to predict the changes. They further describe a method to learn motion parameters from a sequence. Even though good results have been reported, it is not clear how much of the improvement is due to the sophisticated prediction strategy, and how much to the adaptive skin-model idea itself. It should be noted that existing adaptive approaches attempt to explore the relative similarities and differences between the current and the previous frames. Therefore, such techniques can only be applied to videos, not still images. In [2], a novel adaptive approach is proposed for the application of hand-segmentation for arbitrary colors. A restricted EM algorithm is introduced to train an adaptive GMM for still images; wherein the background is modeled by four Gaussian kernels and the hand color is modeled by one Gaussian. This modified EM algorithm requires strict prior information, including a good estimation of hand color and a reasonable bound of weighted values, in order to obtain more robust results. To distinguish the skin component from other Gaussian components, the authors heuristically fix the mean of the first Gaussian in estimating the hand color during the training process. Obviously, this significantly degrades the ability of the GMM to model the actual skin color distribution for individual images.

Skin detection is a key component in the application of automatic detection and filtering of pornographic images. In [7], the system marks skin-like pixels using combined color and texture properties. The marked skin regions are then fed to a specialized grouper, which attempts to group a human figure using geometric constraints on human structure. The efficiency of such a bottom-up framework strongly depends on the accuracy of identifying the skin-like pixels in the initial step. In [8, 9], the authors extract a set of features for characterizing an image, which include color histograms, various

color moments and texture features in different orientations and resolutions. Based on these features, a classifier is trained and used to determine whether an image is pornographic or not. For this image filter, even though the number of features extracted from the skin pixels accounts for less than 10% of the total number of features, it has been observed that these skin-related features are the most critical ones. Therefore, higher accuracy in skin detection should help improve the accuracy of pornographic image detection.

In [12], we proposed the concepts of deriving a unified, adaptive skin-color model for arbitrary still images. In this paper, we extend the initial ideas and refine the techniques, as well as conducting thorough experiments to make a comprehensive comparison between the proposed adaptive skin-model and the current generic skin-models. In addition, we incorporated the proposed model into an objectionable image filtering system ****[9]. By extensive experiments and analysis, we observed the strong correlation between the skin-detection accuracy in the pixel level and the pornographic-image-classification accuracy in the image level. The results on large image databases clearly indicate that, by intelligently integrating the proposed new skin-model, ****'s accuracy can be improved. We further propose a two-level classification procedure using a new bagging scheme in which the classifiers are trained by a mixture of the generic skin-model and the new adaptive skin-model. Experimental results show that this scheme further improves the classification accuracy.

The remainder of our paper is organized as follows. Section 2 presents the general idea of the proposed EM-based adaptive modeling. In Section 3, a comparison of our approach with other skin-detection methods is conducted based on a comprehensive image database. Section 4 summarizes its application to objectionable image filtering. In the last Section, we conclude with a short discussion.

2. TWO-STEP ADAPTIVE MODELING

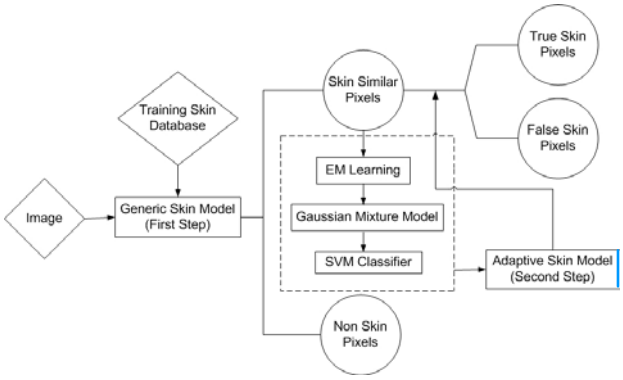


Figure 1: Two-step adaptive framework for skin detection

In contrast to other techniques, our adaptive model for a given image is constructed using information derived specifically from the image under consideration. For a given image, the adaptive skin-model is built based on a two-step process shown in Figure 1. A generic skin-model, built from a training dataset, is first used to classify the pixels in the given image. The pixel set identified as skin pixels in this step is designated as skin-similar space. The goal of the second step is to reduce the false-positive rate without compromising much on the detection rate. We note that our proposed method, in principle, divides the hard task (i.e. analyzing the color distribution of an arbitrary image) into two easier sub-tasks/steps. We believe this work is the first attempt to refine the generic skin-model using such an adaptive, two-step process for arbitrary still images.

2.1 Skin-Similar Space Analysis

We have observed several interesting characteristics inherent in this skin-similar space derived in the first step:

- For individual images, the true skin color distribution within this skin-similar space is pretty consistent. A dominant Gaussian can typically be found in this space, which will be interpreted later as the skin-kernel in a trained GMM.
- A large fraction of the false skin pixels in the skin-similar space usually belongs to the same objects in the image background (e.g. pink wallpaper, curtains with skin-similar colors, etc.). These pixels often form several weak Gaussians. We will explore different numbers of Gaussians (1, 3 and 5) to model the false-skin pixels in our experiments.
- We observed that for most images in the large collection of test images gathered from the web, the true-skin Gaussian and other false-skin Gaussians appear to be quite separable.

One key observation is already apparent: in this simplified, skin-similar space, the GMM with a small number of Gaussian components is generally sufficient to accurately model the real color distribution because the pixels in a noisy background and/or in obvious non-skin regions have been removed from consideration. In the first step, a high detection rate, say over 96%, is needed to ensure that almost all skin pixels are captured (inevitably a high false-positive rate occurs in this step). Therefore, in Step 2, the pixels in the skin-similar space are used to train a GMM with several Gaussian kernels (one for modeling the true skin pixels and the others for modeling the false skin pixels) using the standard EM algorithm. Then an SVM classifier, using the spatial and shape features along with the Gaussian parameters, is proposed to identify the correct skin Gaussian component, i.e. our adaptive skin-model, from the trained GMM.

In [12], we have verified that, for most images, a compact Gaussian (i.e., one with a small variance matrix) can fit the true-skin distribution in the skin-similar space. We have also shown the potential room for reducing the amount of false-skin pixels by plotting the "theoretical lower bound" curve based on the analysis of the overlap between two histograms (skin pixels and non-skin pixels) derived from the ground truth. Our recent investigation concludes that one Gaussian component should be sufficient to fit the distribution of the false skin pixels in the skin-similar Space for application to which the requirement for the detection rate of the true skin pixels is high. In the experimental section, we will present a comparison of the accuracy and the computation efficiency by using different numbers of Gaussian components to model the false-skin pixels in the skin-similar space.

For the sake of clarity, in the following discussion we first use a simple case of the GMM that contains two Gaussian components (one for modeling the true-skin pixels and the other for modeling the false-skin pixels) to illustrate the key steps of the algorithm in the following sections. The main stages in the EM algorithm remain the same for adopting different numbers of the Gaussian components in a Gaussian Mixture Model.

2.2 Unified Adaptive Model Learned by EM

A Gaussian Mixture Model is defined as:

$$P_{hs} = \sum_{i=1}^{k} w_i 2\pi^{-1} |\Sigma_i|^{-1/2} e^{(-\lambda_i(h,s)^2)} \quad (1)$$

where $\sum_{i=1}^{k} w_i = 1$ and k equals to 2 in the simple case. $\lambda_i(h,s) = [(h,s) - u] \sum^{-1} [(h,s) - u]^T$ is the Mahalanobis distance.

Ignoring obscure substantiations, the key updated equations for an EM algorithm are given below. More theoretical details of the EM algorithm can be found in [13].

$$w_l^{new} = \frac{1}{N} \sum_{i=1}^{N} p(l/x_i, \theta^g) \quad (2)$$

$$u_l^{new} = \frac{\sum_{i=1}^{N} x_i p(l/x_i, \theta^g)}{\sum_{i=1}^{N} p(l/x_i, \theta^g)} \quad (3)$$

$$\Sigma_l^{new} = \frac{\sum_{i=1}^{N} x_i p(l/x_i, \theta^g)(x_i - u_l^{new})(x_i - u_l^{new})^T}{\sum_{i=1}^{N} p(l/x_i, \theta^g)} \quad (4)$$

where w, u, and Σ are the weight, mean vector, and covariance matrix in the Gaussian Mixture Model. l indicates which component of the GMM and x_i represents one sample. N is the number of total training samples. With the current model parameter θ^g, $p(l/x_i, \theta^g)$ evaluates the probability of sample x_i belonging to Gaussian kernel l.

When the number of training samples is small, EM basically performs an unsupervised clustering task in the data space. If the structure, such as the number of components in the mixture models, is known in advance, the EM, with a good initial guess, could converge to the true model parameters. Note that both a correct structure assumption and a good initial guess are crucial for producing an accurate model using the EM algorithm. Based on the structure assumption of two Gaussian components for the simple case, two issues need to be resolved: (1) how to obtain a good initial guess, and (2) how to distinguish the skin Gaussian from the trained GMM (with two Gaussian components).

We have observed that the two assumed Gaussian components (for true- and false-skin pixels) are more separable at the S value. So, we calculate the marginal probability over the H value in the 2-dimensional HS color space. Basically, we choose two separate peaks from the marginalized curve of S value as the initial mean vector for the two Gaussian components (obviously, for the configuration of multiple Gaussian components, we can choose multiple peaks in this initial step). For the covariance matrix, empirically we set it to a small initial value.

Now with two Gaussian components built by the EM algorithm, the next step is to identify which component in the trained GMM represents the skin Gaussian in the skin-similar space.

2.3 Identification of Skin Gaussian by SVM

To identify the skin Gaussian, some heuristic rules might be applied. For example, the dominant Gaussian, i.e. the component with a larger weight, can be chosen as the skin model or the one with smaller variance (we always expect compact skin distribution in individual images) is preferred. However, such simple strategies will be challenged for a large collection of arbitrary images. As the skin-detection performance could be seriously degraded if the wrong Gaussian is chosen, a more intelligent method is desirable. The intuitive cues to describe a Gaussian kernel in the trained GMM are its weight, mean and variance. Even though these features are useful, the spatial and shape information for the pixel set corresponding to a Gaussian distribution presents additional distinguishable characteristics. For example, skin pixels often form compact shapes and are less likely to appear on the borders of the image, while the non-skin pixels may spread over the whole image. Hence, we can define them and incorporate this information into the process of classifying the right skin Gaussian. In our experiment, seven types of features, falling into two different categories, are defined as follows:

- Group A (color distribution related): weight, Gaussian mean, Gaussian variance.
- Group B (spatial and shape related): spreadness, elongation [14], X-direction and Y-direction histograms.

Now, with these features, we design an efficient classifier to classify each Gaussian component into either skin or non-skin. Our classifier is based on SVM [15] which is a popular and powerful technique for data classification. Basically, SVM projects the feature space into high dimension space to find a hyper-plane, which theoretically best separates samples belonging to different classes. In our experiment, we use LIBSVM[16] which is a library for support vector classification and regression.

We manually labeled the skin regions for 1120 images as our STD database. Then, using a generic skin-model, we generate the skin-similar map for each image. In the skin-similar space, we build two Gaussians one for skin pixels and one for non-skin pixels based on the manually labeled ground truth. In addition, we calculate the other four spatial and shape features (Group B) for each pixel set individually. These 2,240 samples (two Gaussians, skin and non-skin, for each image) are then used to train our SVM classifier. In the training process, some common techniques, such as scaling feature values and cross-validation based model selection, were applied. Following the same feature extraction process, we generated 1,108 samples from our TDSD database (with 554 images). These samples were used to test the trained classifier. Table 1 shows the experimental results.

Feature	Training Accuracy	Testing Accuracy
Group A	91.0188%	87.7256%
Group B	88.9634%	90.9747%
Group A+B	96.8275%	96.5704%

Table 1: SVM Classification Accuracy

In Table 1, we present three groups of experimental results. The first row gives the training accuracy and testing accuracy using only Group A features for training and classification. The training accuracy is referred to the classification accuracy on the 20% cross-validation set of the training data (which is randomly picked and not used in training). The second row shows the corresponding results using only Group B features. In the last row, features of both groups are used in the classifier. Convincingly higher accuracy is observed for both the training set and the testing set. This experiment strongly indicates that, by combining the Gaussian parameters with the spatial and shape features, SVM can accurately identify

the right skin Gaussian from the trained GMM within the skin-similar space.

3. EXPERIMENTS

3.1 Experimental Datasets

In this section, we provide more details about our dataset. We labeled the skin region manually using Photoshop for 2,720 images to obtain a comprehensive skin database, including skin pixels under different light conditions and from different ethnic characteristics. About 151 million skin pixels are collected in our Skin Tone Dataset (STD). As far as we know, our skin dataset is the biggest among all published datasets except for the work done by CRL [10]. Similarly, 1,500 images containing about 448 million non-skin pixels were manually collected as our Non Skin Dataset (NSD). These two datasets were used to train our generic skin and non-skin models. Due to our special interest in applying the results for objectionable image filtering, the test dataset (TDSD) was constructed by the following steps:

1. 16,500 images with a minimum size of 150x150 pixels were randomly picked from the news, sports and entertainment websites on the Internet.

2. We scan these images using ****, a filtering software for objectionable images. 554 suspicious images meeting these criteria were selected for inclusion in our test dataset for skin detection (TDSD).

Table 2 summarizes the three datasets used in our experiments.

Dataset	Images	Skin Pixels	Non-Skin Pixels
STD	2,720	151 million	0
NSD	1,500	0	448 million
TDSD	554	24 million	75 million

Table 2: Experimental Datasets

3.2 Generic Skin-Model Experiments

Many statistical tools are available to model a given training dataset. In this section, three kinds of current popular models are briefly summarized. During experiments, we adopted the popular HSV color space for our experiments. Furthermore, the HSV space was reduced to its HS subspace by ignoring the V component, which contributes very little for the discrimination of skin tone. A thorough comparison was made by applying different generic skin-models to our test database TDSD. The best model among them serves for the first-step in our adaptive framework.

3.2.1 Mathematic Forms of Generic Skin-Model

Histogram Model[10]

A general histogram model can be easily learned from a large training database. Each of the color components is divided into certain bins (32 in our implementation), and each bin counts the frequency with which such a color value occurred in the entire dataset. The histogram counts are then normalized to a discrete probability distribution form:

$$P_{hs} = \frac{count_{hs}}{total_{hs}} \tag{5}$$

where the numerator represents the count in the corresponding histogram bin, and the denominator is the total counts over all histogram bins.

Single Gaussian Model[11]

For a large group of skin pixels, we can approximate the skin distribution using a bivariate Gaussian model. The probability density function is defined as:

$$P_{hs} = 2\pi^{-1}|\Sigma|^{-1/2}e^{(-\lambda(h,s)^2)} \tag{6}$$

We analytically estimate the mean vector and the covariance matrix using our training database.

Gaussian Mixture Model[11]

The most popular tool of modeling skin distribution is the Gaussian Mixture Model. One advantage of the GMM is its ability to model more distribution details than the Single Gaussian Model. For the real skin distribution, the SGM captures only the biggest peak, whereas the GMM could model multiple peaks by adopting several Gaussian kernels. The general definition of the Gaussian Mixture Model has already been presented in Section 2.2. We choose five Gaussian kernels to fit the skin distribution, i.e. k equals to five in Equation 1.

3.2.2 Skin Classification Using Generic Model

Once we have a model, the simplest classifier can be constructed by evaluating the probability of each pixel, and then comparing it with a given threshold to make a judgment. After introducing the Bayesian rule, a more sophisticated manner is defined as:

$$P(s/x) = \frac{P(s)P(x/s)}{P(s)P(x/s) + P(ns)P(x/ns)} \tag{7}$$

where x is the color value for a given pixel, $P(x/s)$ and $P(x/ns)$ are the skin and non-skin models derived from our training database. The values of $P(s)$ and $P(ns)$ are empirically assigned based on the prior information in the training dataset.

3.2.3 Experimental Results and Discussions

For the convenience of comparing the performance among different models, we apply them to the same test dataset (TDSD), which contains about 24 million skin pixels and 75 million non-skin pixels in 554 images. Classifier performance can be quantified by calculating the ROC curve which measures the trade-off between the detection rate and the false positive rate by adjusting the threshold. In Figure reffig:roc1, we plot three ROC curves to make a clear comparison. We observed that a GMM curve, by and large, outperforms the others. This result confirms the finding that GMM models more distribution details than SGM. Surprisingly, as the simplest model, histogram curve works pretty well. This finding strongly confirms the conclusion given in [10]: using a large dataset, the simple learning rules, such as histogram density estimators, may give good performance.

3.3 Adaptive Skin-Model Experiments

In this section, we mainly report two individual parts. First, we will compare the accuracy and the computation efficiency by adopting different Gaussian numbers to model the false-skin pixels in the skin-similar space. In the second part, a

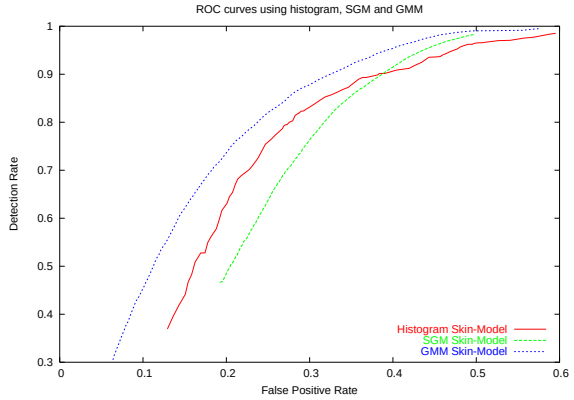


Figure 2: ROC curves using different generic skin-model

comparison between the best generic skin-model (GMM with five kernels) and our adaptive skin-model will be presented in detail.

3.3.1 *Varied Gaussian Numbers for False-skin*

We have observed that the false skin pixels in the skin-similar space of a given image usually belong to one to several objects in the image background (e.g. pink wallpaper, curtains with skin-similar colors, etc.). These pixels often form several weak Gaussians. In this experiment, we explore different Gaussian numbers (1, 3 and 5) to model the false-skin pixels in the skin-similar space. The final performance is shown in Figure 3.

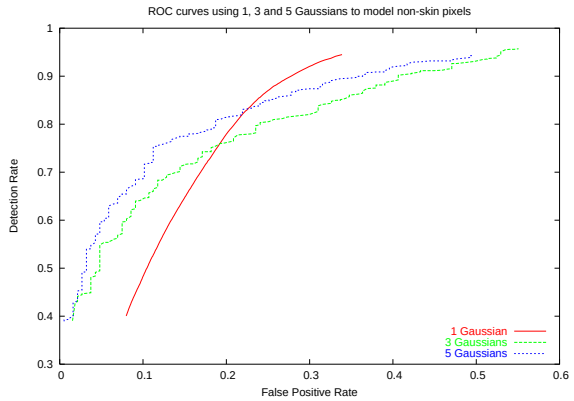


Figure 3: ROC curves using different Gaussian numbers

In Figure 3, we plot three ROC curves to make a clear comparison of adopting different numbers of Gaussian components for modeling the false-skin pixels in the skin-similar space. The red curve, using one Gaussian to model the false-skin pixels, significantly outperforms the others at high detection rates, whereas it degrades quickly at a lower detection rate. The reasons for this result are twofold. First, for the false skin pixels, multiple Gaussian components fit the real distribution better. Therefore, we should obtain better performance by using more than one Gaussian if the Gaussian representing the true-skin is correctly identified. However, the more the Gaussian components we use, the higher the error rate in the step of distinguishing the skin Gaussian from the trained GMM. Inevitably a higher possibility of choosing the wrong Gaussian occurs in this step. At a higher detection rate, we need to set a more stringent threshold, which implies a higher penalty if we choose the wrong Gaussian. Therefore, the configuration with one Gaussian component (for false skin pixels) performs better at a high detection rate. Note that the curves for three-Gaussian configuration and five-Gaussian configuration are quite similar. For most practical applications, a high detection accuracy (>90%) is often required. Moreover, the computation time is roughly a linear function of the number of Gaussian components. Therefore, we choose the one-Gaussian configuration in our later experiments.

3.3.2 *Generic Skin-Model vs. Adaptive Skin-Model*

Three ROC curves are plotted in Figure 4. The blue, dotted curve represents the generic GMM skin-model performance as discussed in Section 3.2.3. The green, solid curve indicates the performance of our adaptive GMM model using the heuristic rules to identify the skin Gaussian. By adopting an SVM classifier to identify the skin Gaussian, a new red, solid curve is plotted for our adaptive GMM model. We noticed that both adaptive models outperform the original generic GMM model. This implies that our method can effectively separate the separable skin and non-skin pixels in the skin-similar space. In addition, the SVM mechanism indeed improves the overall detection performance using only the simple heuristic rules. For example, at 90% detection rate, we reduce the FP rate by about 5.1% (from 32.7% down to 27.6%) using the adaptive GMM1, and about 10.4% (from 32.7% down to 22.3%) by incorporating the SVM classifier in adaptive GMM2.

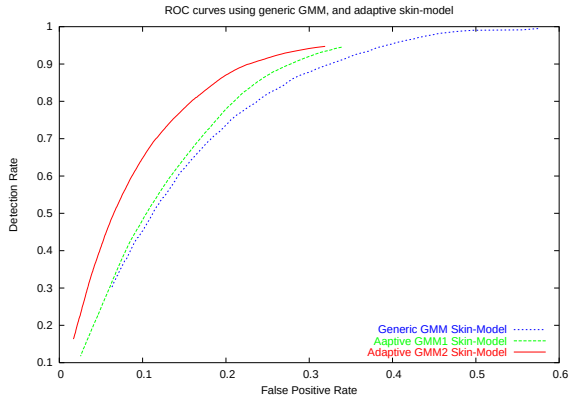


Figure 4: ROC curves using generic and adaptive skin-model

4. OBJECTIONABLE IMAGE FILTERING

Objectionable information is often undesirable, offensive, and sometimes harmful, especially to young children. While Internet access has opened up enormous new possibilities, it has also dramatically increased the risk for end-users to receive objectionable information. The proliferation of objectionable information, e.g. pornography, on the Internet has reached a level of serious concern and there is a growing demand of filtering it.

****[9] is a multimodal personalized information system to empower end-users to screen out objectionable information. ****employs classifiers of multiple layers and multiple modalities to combine information from different types of contents. One of ****'s core components is its objectionable image classifier, which uses Support Vector Machine (SVM) as the base classifier. In ****'s objectionable image classifier, each image in the training and testing set is represented by a feature vec-

tor of 144 features. These include color histograms, means, variances, elongation, and spreadness extracted from 12 natural color channels and texture features in three orientations (vertical, horizontal, and diagonal) extracted at three different resolutions (coarse, medium, and fine resolution)[14, 17]. In addition to the 12 natural color channels, skin pixel information is used as the 13th color channel. The color histogram, mean, variance, elongation, and spreadness of the skin pixels are extracted for both training and classification.

To set up our following experiments, we randomly selected 15,000 objectionable images as the positive examples, and 15,000 benign images as the negative examples, from a set of images crawled from the Internet. Their classes (objectionable/benign) were manually labeled. We further picked 10% (3,000 images) of the image dataset to be our cross-validation test set. The remaining 90% of the dataset (27,000 images) were used as the training set. In this section, we present experiments to study the different impacts of the proposed adaptive skin-model on positive and negative images in image classification. We show how the proposed adaptive skin-model can help improve the accuracy of objectionable image filtering. Finally, we propose a novel idea of hierarchical bagging by using multiple classifiers to further improve the accuracy of objectionable image classifier.

4.1 Improvement by Adaptive Skin-Model

In this experiment, we replaced the original generic skin-model (GSM) by the proposed adaptive skin-model (ASM) in our objectionable image classifier. As the proposed ASM achieves a significantly higher accuracy in detecting the skin pixels, we should expect a higher accuracy for classifying a pornographic image. Table 3 shows the accuracy of the image classifiers using the GSM (C1) and the ASM (C2). The ROC curves using the GSM and the ASM will be shown and discussed later.

Type	Detection Rate / False Positive Rate (%)
C1	91.7/8.29
C2	92.75/7.19

C1: classifier using GSM
C2: classifier using ASM

Table 3: Accuracy using GSM and ASM

4.2 Asymmetric Skin-Models Experiments

We noticed that the ASM has quite different impacts on positive (pornographic) images and negative (benign) images due to their distinct characteristics. For instance, negative images in general have fewer skin pixels than positive images. In addition, skin pixels in negative images are, in general, more dispersed than those in positive images. To further understand and explore the effects of the ASM on image classification, we conducted experiments using an "asymmetric skin-model". In these experiments, we trained two classifiers (C3 and C4) by applying the ASM to either positive or negative training examples, but not to both sides. Specifically, C3 was trained by applying the GSM on positive examples and the ASM on negative examples. Classifier C4, on the other hand, was trained by applying the ASM on positive examples and the GSM on negative examples. In the testing stage, we ran two experiments, Test 1 and Test 2, separately. In Test 1, positive testing images use the GSM whereas the negative testing images use the ASM. In Test 2, positive testing images use the ASM whereas the negative images use the GSM . Table 4 shows the accuracy comparison of the two tests against these two asymmetric classifiers C3 and C4.

As we do not know beforehand whether a testing image is positive or negative, this asymmetric skin-model experiment is not practical in a real application. The experiment is mainly for the purpose of gaining more insights to the core value of the skin-models. Two interesting findings can be derived from the results of these experiments. First, each asymmetric classifier achieves a high detection rate and a low false positive rate when the positive (negative) testing images use the same skin-model as the positive (negative) training images in training stage (entries of C3-Test 1 and C4-Test 2 in Table 4). Second, a significantly lower detection rate and higher false positive rate result, if the positive (negative) testing images use the skin-model opposite to the one uses in the positive (negative) training images (entries of C3-Test 2 and C4-Test 1).

Type	Detection Rate / False Positive Rate (%)	
	Test 1	Test 2
C3	94.67/5.33	75.63/24.53
C4	76.76/23.35	93.43/6.61

C3: training - positive using GSM / negative using ASM
C4: training - positive using ASM / negative using GSM
Test 1: testing - positive using GSM / negative using ASM
Test 2: testing - positive using ASM / negative using GSM

Table 4: Accuracy using Asymmetric Skin-Model

Using the new ASM to extract skin-regions from a given image, the color variance of the skin-pixels may reduce and the extracted skin-regions may become more compact spatially for both positive and negative images. Therefore, our ASM could make some skin-related features, originally separable using the GSM, less distinguishable. Probably, this is why, in Table 3, the improvement achieved by the image classifier using the ASM (C2) over the original image classifier using the GSM (C1) is not as significant as we expected. On the other hand, if we apply the ASM to either positive or negative data alone, we may increase the separability of some features. For example, applying the ASM to the negative examples may further reduce the number of identified skin pixels for negative examples. Therefore, it would increase the difference in skin-pixel ratio between a pornographic image and a benign image. Also, applying the ASM to the positive images makes the skin-pixel regions even more spatially compact than those in the negative images. In summary, asymmetric classifiers may increase the distances in certain skin-related features between a positive/pornographic and a negative/benign image.

4.3 Hierarchical Bagging

Based on the observations derived from the results of the "asymmetric" experiments described above, we propose a novel bagging technique for pornographic image classification using the proposed ASM. This technique differs from the traditional bagging in two ways. First, multiple classifiers are trained using the same training set but with different features. Second, the classification procedure based on these classifiers is divided into two levels, and the final decision is made in a hierarchical fashion. Specifically, we use an ordinary classifier based on ASM (C2) as the first level classifier, and then use

the two asymmetrical classifiers (C3 and C4) in the second level.

The hierarchical classification works as follows: Each test image is first classified using the level-one classifier. If its SVM score falls within a pre-specified, bounded range (called inner region in Figure 5), the two level-two classifiers will be further used. The final decision will be made based on the majority vote of the three classifiers (C2, C3 and C4). On the other hand, if the SVM score is larger than the upper bound of the specified range or smaller than the lower bound (i.e. falls in the outer range in Figure 5), we make the final classification completely based on this score.

The score reported by an SVM classifier indicates the distance of the test sample to the SVM boundary. The closer the SVM score is to the boundary, the lower the confidence the SVM score provides. If we completely rely on the level-one classifier for class prediction (i.e. classifying the test sample as positive if the SVM score is larger than the boundary, and vice versa), the test samples with an SVM score closer to the boundary have a higher probability to be mis-classified than those with a score farther away from the boundary. Therefore, we set an upper bound and a lower bound on the SVM score of the level-one classifier to determine whether bagging will be used or not.

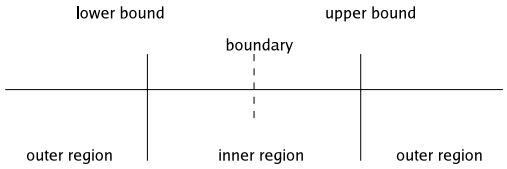


Figure 5: Inner and Outer Regions

By adjusting the upper and the lower bounds, we get variances in the classification accuracy. We conducted some analysis by adjusting the bounds based on the ratio of the test images falling within the inner region to the total test images. Table 5 shows the results of this analysis.

ratio	outer acc.	inner acc.	theoretical acc.
20%	94.40%	87.70%	93.77%
30%	95.25%	86.92%	94.19%
40%	96.38%	87.31%	94.54%
50%	97.31%	88.19%	94.63%
60%	97.70%	89.45%	94.38%

Table 5: Theoretical accuracy analysis using hierarchical bagging

The first column of Table 5 shows the ratio: the number of testing images with an SVM score (from classifier C2) falls within the inner region to the total number of testing images. The second and third columns are the classification accuracy (of classifier C2) with respect to the subset of testing images falling within the outer region and the inner region respectively. In the hierarchical classification scheme, the accuracy for the testing images falling within the outer range remains the same, while the accuracy for the images falling within the inner range will be improved for the incorporation of the asymmetric classifiers C3 and C4 in the second level. The final accuracy of the entire testing set is equal to the weighted sum of the original "outer accuracy" and the refined "inner accuracy". Table 5 shows the theoretical accuracy in the fourth column based on Equations 8 and 9. As indicated, about 2% further improvement, beyond that which the original C2 classifier provides, can be achieved theoretically:

$$P = ratio * P_i^r + (1 - ratio) * P_o \quad (8)$$

$$P_i^r = P_i \frac{P(C_{31}) + P(C_{42})}{2} + (1 - P_i) \frac{P(C_{32}) + P(C_{41})}{2} \quad (9)$$

where P_i and P_o represent the classification accuracy of classifier C2 in the inner range and the outer range shown in Table 5. $P(C_{31})$ is the classification accuracy of classifier C3 assuming the "Test 1" process described in Section 4.2 is employed (i.e. entry of C3-Test 1 in Table 4). Similarly, $P(C_{42})$, $P(C_{32})$, and $P(C_{41})$ correspond to the entries of C4-Test 2, C3-Test 2 and C4-Test 1 in Table 4 respectively.

In Figure 6, we show the experimental ROC curves for the 3,000 testing images using the proposed hierarchical classification scheme. 5 curves are plotted: each of which represents a different bound corresponding to each of the five cases in Table 5. We notice that the real performance is surprisingly consistent with the theoretical analysis given above. When the ratio is equal to 50%, the accuracy seems to be the highest. The accuracy is lower for cases of ratio at 20% and 30% (green and blue dotted curves).

In Figure 7, we show three ROC curves comparing the final performance of three classification schemes. The red solid curve indicates the performance of the original classifier C1. The green dotted curve shows the performance of classifier C2 which uses the adaptive skin-model for skin-related features. C2 curve is superior to C1 curve, confirming the value of the new skin-model to this application. The blue dotted curve, produced using the new hierarchical bagging scheme, clearly shows the best performance. Table 6 compares the classification accuracy of these three classification schemes by choosing the neutral point on the ROC curves shown in Figure 7. The neutral point is the one that the false negative and the false positive rates are about the same.

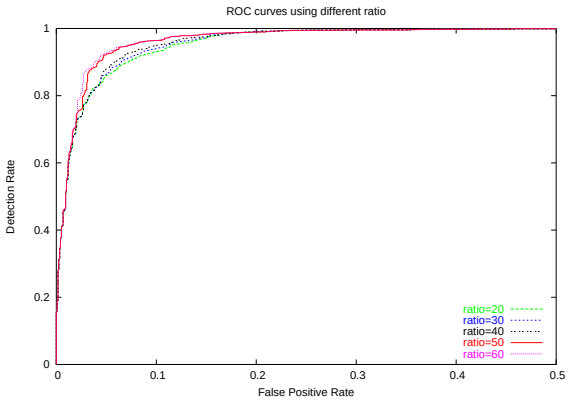


Figure 6: ROC curves using different ratio

Type	Detection Rate / False Positive Rate (%)
C1	91.7/8.29
C2	92.75/7.19
C-HB	94.0/6.01

Table 6: Accuracy using GSM, ASM and hierarchical bagging

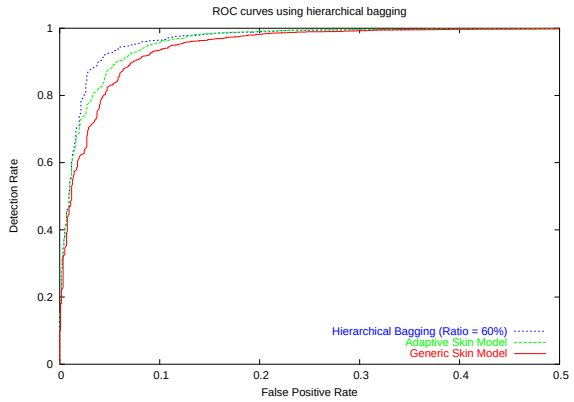


Figure 7: ROC curves using hierarchical bagging

4.4 Computation Time Analysis

Type	Feature	Classification	Overall
C1	129.81ms	60.47ms	190.26ms
C2	216.15ms	61.29ms	277.44ms
C3	216.15ms	50.41ms	266.59ms
C4	216.15ms	50.15ms	266.30ms
C-HB(60%)	216.15ms	84.852ms	301.00ms

Table 7: Computation Time Analysis (per image)

In this section, we present CPU runtime on feature extraction and on image classification. The reported runtime for feature extraction includes the extraction of all 144 features including the 9 skin-related features. On the average for the 3,000 testing images, classifiers C2, C3, C4 and C-HB (using the ASM in this step) take about 66% additional runtime per image in comparison with classifier C1, which uses the GSM. In the classification phase, C1, C2, C3 and C4 take roughly the same amount of time to classify an image. The slight difference shown in Column 3 of Table 7 is mainly caused by the different numbers of support vectors for these four classifiers.

For the two-level classifier using the hierarchical bagging, i.e. C-HB, the classification time is longer because, for some images, two more classification runs are needed. Column 4 shows the overall runtime per image, which is the sum of Columns 2 and 3. Classifier C-HB, which achieves the highest accuracy, incurs about 60% additional runtime overhead in comparison to the original classifier C1. As the overall runtime remains very low (about 0.3 second per image), such overhead in computational time should not be a concern for most applications.

5. CONCLUSION

This paper presents a novel two-step adaptive framework for accurate skin-color detection. In the first step, we identify the Skin-Similar pixels using a generic skin-model. The standard EM algorithm is then used to train a GMM. An SVM classifier, using the spatial and shape features along with Gaussian parameters, is trained to distinguish the correct skin Gaussian component from the trained GMM. In comparison with traditional methods which reply on a generic skin-model, the experimental results validate that the new method achieves a significantly lower false-positive rate for skin detection. We have further demonstrated that the proposed model helps improve the detection accuracy of pornographic images. Moreover, we propose a "hierarchical bagging" classification scheme to intelligently utilize the new adaptive skin-model for further improvement in pornographic image classification. Even though the number of features extracted from the skin pixels accounts for less than 10% of the total number of features used in the image filter, the detection accuracy based on the new classification scheme and the new skin-model is significantly higher than that of the original filter which depends on the generic skin-model.

6. REFERENCES

[1] R.-L. Hsu, M. Abdel-Mottaleb, and A.K. Jain. Face detection in Color Images. IEEE Trans. on Pattern Analysis and Machine Intelligence, vol. 24, no. 5, pp. 696-706, May 2002.

[2] X. Zhu, J. Yang, and A. Waibel. Segmenting hands of arbitrary color Proc. IEEE Intl. Conf. on Automatic Face and Gesture Recognition (FG 2000), pp. 446-453, Mar. 2000, Grenoble, France.

[3] C. Garcia and G. Tziritas. Face Detection Using Quantized Skin Color Regions Merging and Wavelet Packet Analysis IEEE Trans. on Multimedia, vol. 1(3): pp. 264-277, Sept. 1999.

[4] Y. Wu and T.S. Huang. Nonstationary Color Tracking for Vision-Based HumanComputer Interaction. IEEE Trans. on Neural Networks, vol. 13, pp. 948-960, July 2002.

[5] J. Yang, W. Lu, and A. Waibel. Skin-Color Modeling and Adaptation. Proc. of the 3rd Asian Conference on Computer Vision (ACCV98), Vol. 2, pp. 687-694, Jan. 1998, Hong Kong.

[6] L. Sigal, S. Sclaroff, and V. Athitsos. Estimation and Prediction of Evolving Color Distributions for Skin Segmentation under Varying Illumination. Proc. IEEE Computer Vision and Pattern Recognition. (CVPR 2000), pp. 152-159, June 2000, CA, USA.

[7] D.A. Forsyth and M.M. Fleck. Automatic Detection of Human Nudes. Int. Jour. of Computer Vision, Vol. 32 , No. 1, pp. 63-77, Aug., 1999.

[8] J. Wang, J. Li, G. Wiederhold and O. Firschein. System for Screening Objectionable Images Using Daubechies Wavelets and Color Histograms. Proc. Interactive Distributed Multimedia Systems (IDMS'97), Volume 1309, Springer-Verlag LNCS, pp. 20-30, Sept. 1997, Darmstadt, Germany.

[9] Remove for double blind review purpose

[10] M.J. Jones, J.M. Rehg. Statistical Color Models with Application to Skin Detection. Technical Report, Cambridge Research Lab., 1998.

[11] J.C. Terrillon, M.N. Shirazi, H. Fukamachi, and S. Akamatsu. Comparative Performance of Different Skin Chrominance Models and Chrominance Spaces for the Automatic Detection of Human Faces in Color Images. Proc. IEEE Intl. Conf. on Face and Gesture Recognition (FG 2000), pp. 54-61, Mar. 2000, Grenoble, France

[12] Remove for double blind review purpose

[13] J. Bilmes. A Gentle Tutorial of the EM Algorithm and its Application to Parameter Estimation for Gaussian Mixture and Hidden Markov Models. Technical Report 97-021, Intl. Computer Science Institute, Univ. of California, Berkeley, Apr. 1998.

[14] Remove for double blind review purpose

[15] C.J.C. Burges (1998). A Tutorial on Support Vector Machines for Pattern Recognition. Kluwer Academic Publishers, Boston.

[16] C.-C. Chang and C.-J. Lin. LIBSVM: a library for support vector machines, 2001. Software available at http://www.csie.ntu.edu.tw/ cjlin/libsvm

[17] Remove for double blind review purpose